

IoTeX: Building a connected world powered by IoT and blockchain with Google Cloud solutions

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ABOUT IOTEX



With its blockchain platform, IoTeX is handling 35 million transactions for IoT devices, and people every year with 99.9% reliability, achieving month-on-month growth rates via Google Cloud.

About IoTeX

IoTeX is an open source platform at the intersection of blockchain and the Internet of Things, on a mission to enable the future decentralized machine economy. It seeks to build a connected world where machines, humans, businesses, and decentralized applications (DApps) can interact with trust and privacy. Backed by a global team of 40+ research scientists and engineers, IoTeX combines blockchain, off-chain compute, and open hardware to connect billions of devices and DApps across the physical and digital world.

Google Cloud results

- Builds trust in blockchain IoT ecosystem with 99.9% reliability rate and zero system failures
- Empowers developers and users with real-time IoT transactions handling one million API calls per day
- Supports 20% month-on-month growth rate with unlimited compute power and seamless autoscaling of Google

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million
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Industries: Financial Services & Insurance

Location: Singapore

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>),
Google Cloud (<https://cloud.google.com/>)

Compute Engine

(<https://cloud.google.com/compute/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Tell us your challenge. We'll have to help

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Contact us (/contact)

Confidential Computing

(<https://cloud.google.com/confidential-computing>)

Kubernetes

Engine

Autopilot

- Enables ultimate security on blockchain back end with advanced Confidential Computing solution

The Internet of Things (IoT) is predicted to drive the fourth industrial revolution. According to McKinsey (<https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/iot-value-set-to-accelerate-through-2030-where-and-how-to-capture-it>)

, the economic value generated by IoT is expected to nearly double to \$12.6 trillion by 2030, with separate research

(<https://www.statista.com/statistics/1183457/iot-connected-devices-worldwide/>)

projecting the number of connected devices to grow 150% from 10 billion in 2021 to 25 billion in the same year.

This remarkable value creation promises to transform just about every aspect of life, from healthcare to clean energy and driverless mobility. For the creators of IoTeX (<https://iotex.io/>), an open source platform at the intersection of IoT and blockchain, it's crucial to ensure that the opportunities unleashed by the IoT revolution are

open to all, not only big players with control over data that are the lifeblood of connected device networks.

"If data is the new gold, then IoT-linked machines will be the gold mines.

We believe devices will be the biggest users of blockchain in the future. Our mission is to give people control over these assets.

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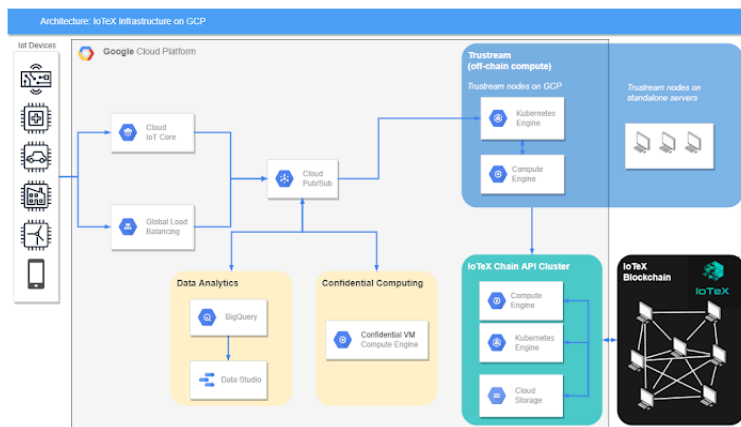
*—Raullen Chai, CEO and
co-founder, IoTeX*

Calling its vision MachineFi, IoTeX pulls data from potentially billions of IoT devices around the world onto the blockchain, creating a verifiable "single version of the truth" for the assets, as the foundation of trust. The implications of the innovation are significant. It means Web3 developers can engage at a peer-to-peer (P2P) level with end users of devices, building incentive-based value creation, without the need for intermediaries. This in turn enables a future decentralized real-world economy, in the same way decentralized finance (DeFi) enables a virtual crypto economy.

To succeed, IoTeX needs four critical cloud computing factors. Rock-solid security, seamless scalability, world-class data crunching, and a zero-latency environment. By using [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>), IoTeX discovered the tools it needed to enable imaginative developers around the world to build a new economic paradigm, merging the familiar world of physical things with the blockchain revolution.

"If data is the new gold, then IoT-linked machines will be the gold mines. We believe devices will be the biggest users of blockchain in the future. Our mission is to give people control over these assets," says Raullen Chai, CEO and co-founder, IoTeX. "With a combination of Google Kubernetes Engine Autopilot

(</kubernetes-engine/docs/concepts/autopilot-overview>) (GKE), BigQuery (</bigquery>), and Cloud Load Balancing (</load-balancing>), we can turn this vision into reality.



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universe. In the future, the IoT ecosystem platform will secure billions of connected devices feeding their data snapshot to the blockchain. With GKE Autopilot and Cloud Load Balancing, we can easily absorb any load no matter how much or how fast we grow."

—Larry Pang, Head of Ecosystem, IoT ecosystem

Unleashing decentralized IoT co-creation with GKE Autopilot, BigQuery, and Cloud Load Balancing

Imagine being an insurtech startup inventing new ways to empower people with AI-driven risk management tools. In the traditional insurance

model, dominant players might lower car insurance premiums for every year the customer has not been in an accident. But does that make the customer a safe driver? .

The world today operates on penalties for doing things wrong, but what if insurtech could incentivize customers to drive safely by collecting IoT data from their vehicle, to reward them for wearing a seatbelt and always using turn signals? The rewards might be lower premiums, but they could also be crypto tokens or any other form of digital value exchange.

This is one of innumerable potential scenarios in the IoTeX "Web 3.0" vision of the decentralized economy, powered by IoT, underwritten by blockchain, and hosted on Google Cloud. IoTeX aims to extend incentives from the digital world to the physical world, where machines are the arbiters of truth, creating win-win scenarios. Other use cases might be a green NGO that tracks home energy data, giving tokens for eco-friendly practices, or a travel guide that rewards tourists for completing custom itineraries using "proof of presence," transmitted by a device, to verify they have visited a list of featured destinations.

Google Cloud supports IoTeX by enabling two critical conditions for the solution's performance: verification of data and availability of data. The first condition, verification, is important because the data sent into the registry might be falsified or corrupted at origin, even though the blockchain itself enables trustless proof of ownership.

IoTeX has its own decentralized solution for both verification and availability, called TruStream (<https://docs.iotex.io/layer2/trustream>). This decentralized protocol programmably pulls verifiable, real-time data from open devices into the IoTeX network. Here, availability of data becomes crucial to the real-time, zero-latency automated interaction at the core of the platform. At any given time, IoTeX must be able to handle a virtually unlimited number of API calls to the IoTeX blockchain, where the data is stored.

The use of BigQuery on top of TruStream and cross-referencing with OpenStreetMap (<https://www.openstreetmap.org/>), enriches the machine or IoT data. Data is also easily visualized with Looker Studio (<https://marketingplatform.google.com/about/data-studio/>) and easier for developers to use. IoTeX leverages the vast library of public datasets ([/bigquery/public-data](#)) hosted on BigQuery, before indexing the metadata of machines registered on the IoTeX blockchain. This process enables developers to understand the chain identity of machines that feed data into the platform.

The backbone of this turbocharged datastream is GKE Autopilot, a new mode of operation in Google Kubernetes Engine. The containerized infrastructure solution provides seamless autoscaling to handle any spike in demand from the global IoTeX device ecosystem. Meanwhile, Cloud Load Balancing enables the zero-latency connections critical for

real-time IoT transaction, by linking machines to the data center closest to the origin of data.

Thanks to Google Cloud, says Chai, IoTeX has achieved the ultimate platform reliability and real-time data processing it needs to power a global, decentralized IoT network. Since launching in 2019, IoTeX has handled more than 35 million transactions, without a single system failure, for a 99.9% reliability rate, running on Google Cloud. Every day, IoTeX handles more than 1 million API calls for data to its blockchain, thanks to the seamless autoscaling of GKE Autopilot and Cloud Load Balancing.

"We regularly experience massive scaling surges from random places in the crypto universe. In the future, the IoTeX platform will secure billions of connected devices feeding their data snapshot to the blockchain. With GKE Autopilot and Cloud Load Balancing we can easily absorb any load no matter how much or how fast we grow," says Larry Pang, Head of Ecosystem at IoTeX.

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expansion of Google Cloud deployment for servers, products, and services. One exciting project is using machine learning to verify data from IoT devices. Google Cloud's ML solutions can play a big role, as we write the next chapters of the decentralized IoT adventure."

—Raullen Chai, CEO and co-founder, IoTeX

Impregnable backend security with Google Cloud confidential VMs

IoTeX is experiencing rapid growth. In addition to verification and availability, the business also needs

to meet the challenge of scaling up its API clusters, while ensuring security and reliability in its operations. This is particularly important since the data consists of billions of private data points from devices around the world. "We're confident Google Cloud provides best-in-class security in terms of authentication, authorization, and encryption for data, which is essential for a solution such as ours."

Yet for IoTEx, it also became important to create an extra layer of security for its platform back end, where it keeps its motherlode of IoT data. While IoTEx's user-facing infrastructure runs on GKE for granular and seamless autoscaling, IoTEx runs its own virtual machines (VMs) on Compute Engine (/compute) to index IoT data on its blockchain.

To derive value from the data without compromising the privacy of users and devices, IoTEx is experimenting with a new Google Cloud solution called Confidential Computing (/confidential-computing). Chai says IoTEx learned about confidential VMs while they were still in prototype stage, thanks to a proactive ethos at Google Cloud that goes beyond a client-vendor relationship, to become a true partnership.

"We approached the Google Cloud team about our security needs, and they went beyond the call of duty, sharing the prototype Confidential Computing solution. And together we made it work," says Chai. "It's this teamwork where we co-develop the relationship, and find solutions together, that makes working with Google Cloud so satisfying."

For IoTeX, the partnership is set to evolve in exciting new directions, as the platform pursues a global expansion that unfolds at a blistering pace, running on Google Cloud solutions.

"We're growing 20% month-on-month. That means continuous expansion of Google Cloud deployment for servers, products, and services," says Chai. "One exciting project is using machine learning to verify data from IoT devices. Google Cloud's ML solutions can play a big role, as we write the next chapters of the decentralized IoT journey."

